Problem sheet 8 for Rigid Analytic Geometry Summer Term 2026

Recall that a map $X \xrightarrow{f} Y$ between topological spaces is open if and only if $f(U) = \{f(x) \mid x \in U\} \in \mathfrak{O}_Y$ for all $U \in \mathfrak{O}_X$. Recall also that a morphism of topological rings is a continuous ring homomorphism.

Problem 1 (1 point)**.** *Let $A \xrightarrow{f} B$ be a morphism of topological rings which is open. If $M \subseteq A$ is a bounded subset, show that $f(M)$ is bounded in B!*

Problem 2 (1 point)**.** *Let A be a topological ring and $I \subseteq A$ an ideal. We equip A/I with the quotient topology in which U is open if and only if its preimage in A is. Show that the image in A/I of a bounded subset of A is bounded.*

In the following, the field K is always assumed to come with an ultrametric absolute value $|\cdot|$ for which it is complete and non-discrete.

Problem 3 (2 points)**.** *As in Problem 8 from the previous sheet, let $A = K \oplus K \cdot \varepsilon$ with $\varepsilon^2 = 0$, equipped with the topology of K^2. Let $X = K \cdot \varepsilon$, let $r \in (0,1)_{\mathbb{R}}$*

$$Y = \{t \cdot 1 + 0 \cdot \varepsilon \mid t \in K \text{ and } |t| \leq r\}.$$

Show that Y is topologically nilpotent while $X \cup Y$ fails to be topologically nilpotent.

Remark 1. *Note that by the aforementioned result of Sheet 7, X is also topologically nilpotent. Problem 10 from sheet is thus wrongly posed. All solutions giving either a counterexample or a proof of the following correct assertion will be given the full number of points:*

> *If X and Y are bounded and topologically nilpotent subsets, then $X \cup Y$ is topologically nilpotent.*

The above considerations also provide a counterexample to the stronger assertion where only Y is assumed to be bounded.

Problem 4 (1 point)**.** *Let A be a topological ring, $X = \mathrm{Spa}(A, A^+)$ equipped with what was called the strong topology in the lecture, and $x \in X$. If y is a horizontal specialization of x in $\mathrm{Spv}(A)$, show that $y \in X$ and the specialziation also holds in X.*

The following is Fact 3.2.11 from the lecture with a straightforward proof which is, however, going to be omitted in the lecture. It is crucial in the proof of Tate acyclicity assertions[1] in the case of Huber rings without a topologically nilpotent unit.

[1]Which need additional assumptions to hold!

Problem 5 (1 point)**.** *Let A be a topological ring and $X = \mathrm{Cont}(A)$ or $X = \mathrm{Spa}(A, A^+)$ equipped with what was called the strong topology in the lecture. If x is a closed point of X, show that $\Gamma_x = c\Gamma_x$.*

This can be compared with

Problem 6 (1 point)**.** *In the previous problem, assume that A has a topologically nilpotent unit. Show that $\Gamma_x = c\Gamma_x$ holds for all $x \in X$.*

The following is Fact 3.2.12 from the lecture, the proof which is not going to be given in the lecture although it is short and straightforward.

Problem 7 (2 points)**.** *Let A be a topological ring and $X = \mathrm{Cont}(A)$ or $X = \mathrm{Spa}(A, A^+)$ equipped with what was called the strong topology in the lecture. Let s be specialization of η in X, and let x be a specialization of η in $\mathrm{Spv}A$ such that s is a specialization of x in $\mathrm{Spv}A$. Show that $x \in X$ and the two specialization relations in $\mathrm{Spv}A$ involving x also hold in X!*

Remark 2. *Thus, some results of Huber about the decomposition of specializations in* Spv *which have been omitted from the lecture but not from the script put online during the Pentecost vacations also hold for X, namely Folgerung 1.5.2, Satz 1.5.5 and Bemerkung 1.5.3.*

From now on we will assume K to be algebraically closed. We continue the investigation of van der Put points of $X = \mathrm{Sp}\mathbb{T}_1 = X^\circ$ where $\mathbb{T}_1$ is the one variable Tate algebra over K. Recall the sets of balls $\mathfrak{K}_{<R}$ and $\mathfrak{K}_{\leq R}$ from the previous sheet. In the following, let always $\xi \in X^*$.

Problem 8 (3 points)**.** *For $R \in (0,1]_{\mathbb{R}}$, show that $\xi \cap \mathfrak{K}_{<R}$ and $\xi \cap \mathfrak{K}_{\leq R}$ have at most one element!*

This uniquely determined $< R$-ball or $\leq R$-ball in K° whill be denoted $B_{<R}(\xi)$ or $B_{\leq R}(\xi)$ if it exists. Let M_ξ be the set of all $R \in (0,1]_{\mathbb{R}}$ for which $B_{\leq R}(\xi)$ exists.

Problem 9 (1 point)**.** *We have $1 \in M_\xi$ and $B_{\leq 1}(\xi) = K^\circ$.*

Let R_ξ denote the infimum of the non-empty subset M_ξ of R.

Recall that set $\mathfrak{Rat}_X$ of open subsets of X for which we enforced G_+-quasicompactness by the definition of the Grothendieck topology on $\mathfrak{O}_X$ is the set of open subsets of the form

$$\Omega = \left\{ x \in K^\circ \;\middle|\; |f_0(x)| \geq |f_i(x)| \text{ for } 1 \leq i \leq m \right\}$$

where $\mathbb{T}_1$ is generated by the f_i as an ideal in $\mathbb{T}_1$. This assumption ensures that $f_0(x) \neq 0$ for all elements of this set. Moreover,

Problem 10 (2 points)**.** *In the above situation, show that there is a positive real number ε such that $|f_0(x)| \geq \varepsilon$ holds for all $x \in \Omega$!*

Problem 11 (2 points)**.** *In the above situation, show that Ω is clopen for the ordinary topology on X.*

We can now return to the investigation of our $\xi \in X^*$.

Problem 12 (6 points)**.** *If $R_\xi = 0$, show that there is $x \in X$ such that $\xi = \left\{U \in \mathfrak{O}_X \;\middle|\; x \in U\right\}$!*

Of the 23 points from this sheet, three are bonus points which do not count in the calculation of the 50%-threshold for passing the exercise module.

Solutions should be e-mailed to your tutor s45qhou at uni hyphen bonn dot de before Monday June 22. Good luck!